The Performance of Electronic Veterinary Management

Kongkarn Dullayachai
Computer Science Department
Maejo university
Chiangmai, Thailand
kongkarn@gmaejo.mju.ac.th

Attawit Changkamanon
Computer Science Department
Maejo university
Chiangmai, Thailand
attawit@gmaejo.mju.ac.th

***Abstract*—There are three types of user in the Electronic Veterinary Management (EVM) : employee; veterinarian; and manager. The employee role is the weakest authority and the manager is the strongest. Employees can manage the master data in the system that are the pet's owner, pet data, treatment fee, and the queue system. Veterinarians can modify the treatment data. The final users, the manager can manipulate the user authority data and access the revenue of the clinic. The architecture of this system is model, view, and controller by servlet, JSP and Java language, MySQL is the database management system and iReport is the report generating tool. The average EVM time for getting pet's data is 25.14 seconds and a standard deviation value is 10.02. While as the average time of the original process is 37.09 seconds, s.d. is 23.24. Therefore EVM is faster accessing than the original pet record procedure.**

Keywords—Veterinary management system, Veterinary clinic, Veterinary information system, Pets management system

I. Introduction

Nowadays, the popularity of pet care has expanded from the urban area to the suburbs. So there are many veterinary clinics spread over the wide area. However, most internal clinical management is done manually with paper documents. To decrease the waste cost about this problem, the veterinary information system should be a choice. we developed the Electronic Veterinary Management (EVM) for solving the documental problem and the queue management system in the small veterinary clinic, based on freeware tools.

II. Method

A. Data collection

There are 4 ways to get the requirement data from the veterinary clinic.

1. The veterinary medical forms, they are transferred to electronic forms, e.g. veterinary outpatient form, registration form, pet vaccination chart.
2. The veterinary medical records, they are the raw data.
3. Interviews with the veterinarians and all staff to clarify what they want.
4. Observation of the internal veterinary process.

B. System analysis

The comparison of function and cost between general veterinary information system with CVS.

TABLE I. THE COMPARISON OF FUNCTION & COST

Function and Cost	***EVM***	***General Veterinary Information System***
Pet record	✓	✓
Pet history	✓	✓
Pharmaceutical management & stock	Pharmaceutical management	✓
Appointment	✓	✓
User permission	✓	✓
License fee	-	✓

"Fig. 1", at the left hand flowchart, there are 7 steps in the manual workflow that was used before setting up the information technology system. The followings are the basic steps :

1. Client writes his own details and pet information into the health record.
2. Staff gives the queue number in the line to the client.
3. Client waits to be called.
4. Staff calls for the next queue number to see the veterinarian.
5. Veterinarian asks about the pet symptoms and diagnose.
6. Staff gets the diagnosis details from the veterinarian and calculate the expenses.
7. Client gets the medicines.

When the EMV was used, shown on the right hand flowchart, only 4 activities were replaced by the information technology system : Staff inputs the client's details and pet's information in to the system; veterinarian accesses the next queue number; veterinarian asks about the pet symptoms before recording the result of the diagnosis into the system and the last activity; staff calculates the total expense of the client.

Authorized licensed use limited to: University of Exeter. Downloaded on July 02,2020 at 16:40:58 UTC from IEEE Xplore. Restrictions apply.

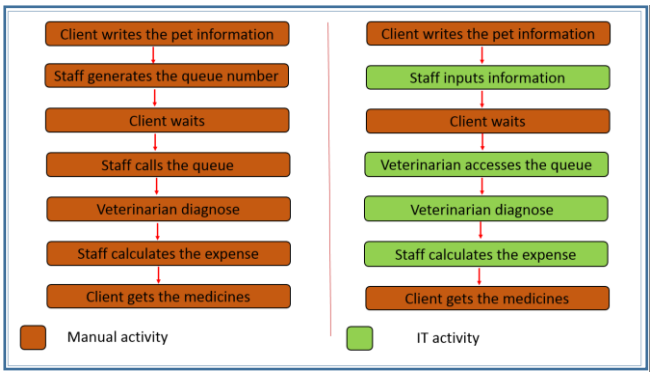

Fig. 1. The manual and IT workflow.

"Fig. 2", the cooperation of 3 users when they use the EVM. The username and password of each user will be verified before giving the authority to access the system. Staff can register for the clients and generate the queue number for them. The veterinarian gets the next client from the queue system, asks them for the pet's symptoms, diagnoses and saves the diagnosis data. The client gets the required medicine for the pet from the staff and pays the expense. The manager can define the authority to each user and look at the summary of incomes.

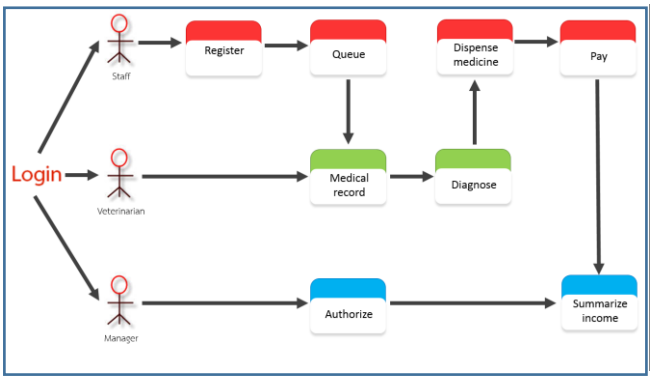

Fig. 2. Cooperation of all users.

The functions of each role are shown in a usecase diagram. "Fig. 3" shows the functions of all users in the EVM system. The staff, the low level user, can manage pet information, queue system, pet health record and client payment system. While as the veterinarian can access only 2 features, getting the pet data in the queue system and adding the pet health after recognizing the pet symptom. The highest user, manager, can manipulate the payment system, retrieve the revenue summary report and define user permission.

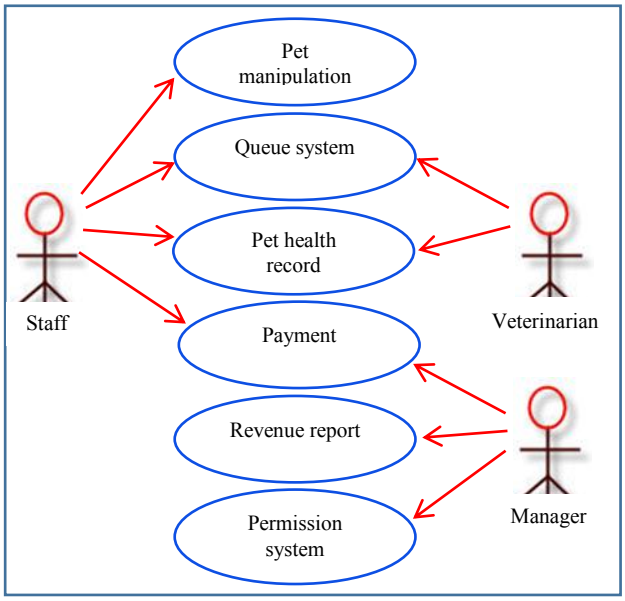

Fig. 3. The user functions.

C. System design

The class diagram of this system is composed of 9 classes, as shown in "Fig.4". These are as follows : Drug, Members, OtherTreatment, Ownerdata, Petdata, Treatment, Treatment_Drug, TreatmentOther, and Type_drug. In the diagram, 8 classes are related to each other but only Members class is not related to the others. The relationships between classes such as Association, Aggregation, Composition etc., are shown in the diagram.

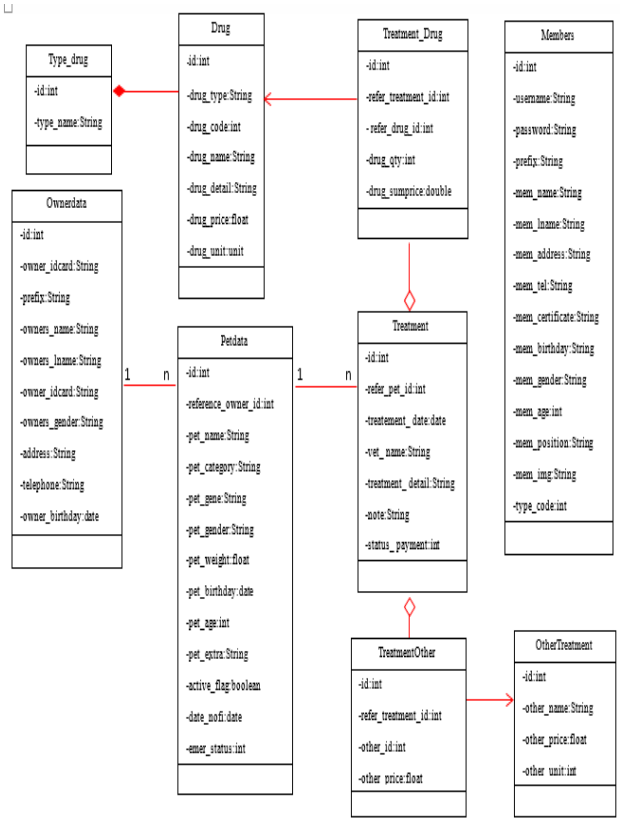

Fig. 4. Class diagram.

Authorized licensed use limited to: University of Exeter. Downloaded on July 02,2020 at 16:40:58 UTC from IEEE Xplore. Restrictions apply.

D. Database design

The database system was designed in accordance with the class diagram, shown in "Fig.5". The EVM's database system classifies 2 types of table, master and transaction. The master tables comprises 6 tables as follows :

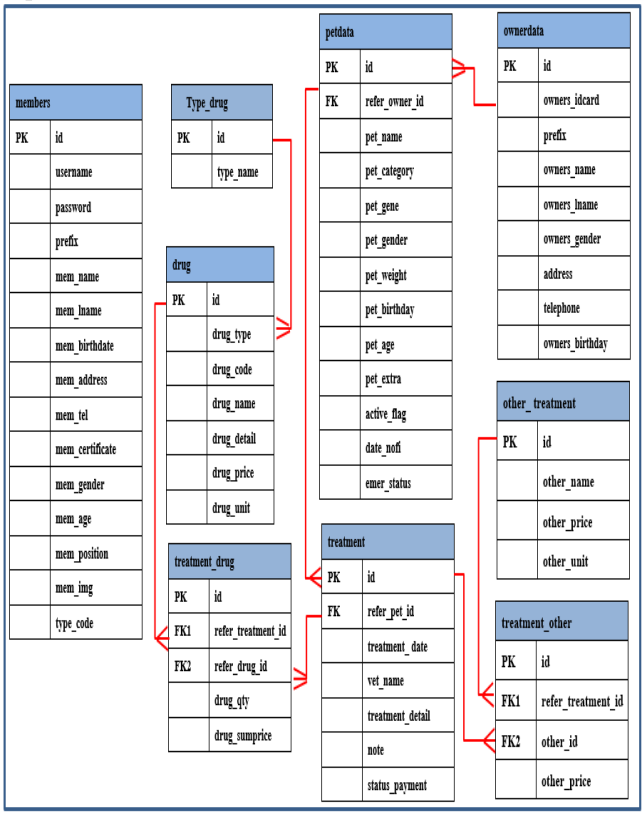

Fig. 5. The veterinary entity relationship diagram.

1. drug : the master table stores all drug details.
2. members : the master table stores all user details, staff, veterinarian, and manager.
3. other_treatment : the master table stores all accessary details except for drug information.
4. ownerdata : the master table stores all client details.
5. petdata : the master table stores all pet details.
6. Type_drug : the master table stores all types of medicine.

There are 3 transaction tables in this database system as follows :

1. treatment : the transaction table stores all treatment and diagnosis details.
2. treatment_drug : the transaction table stores all medicines that the veterinarian ordered.
3. treatment_other : the transaction table stores all accessories that the veterinarian ordered.

E. Interface design

The EVM was allowed to trial by the manager of Watana Rak Sa Sat clinic so this system was designed for that clinic. Each user can access the system with the url, http://202.28.37.31:8080/clinic_web and then login function will verify the user to get into the system.

These are some staff screen examples. "Fig.6" it is the screen of pet detail.

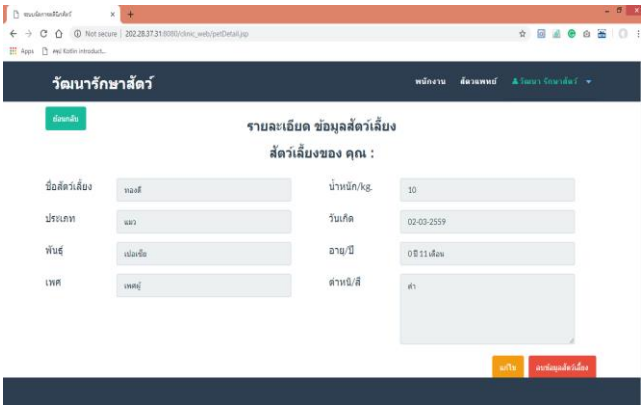

Fig. 6. The pet detail.

The following are some of the veterinarian screens. The veterinarian diagnoses and saves on "Fig.7". Medicines and accessories are dispensed by the veterinarian after he diagnoses the pet disease on "Fig. 8"

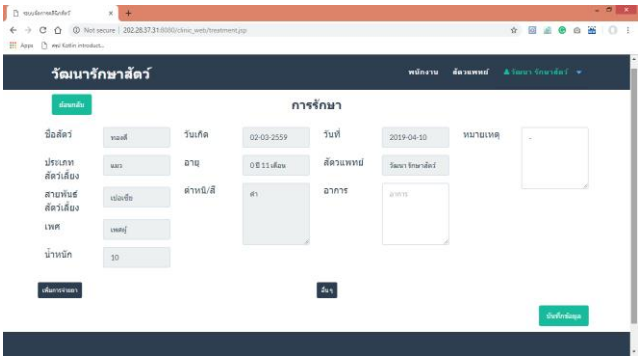

Fig. 7. Veterinarian diagnosis screen.

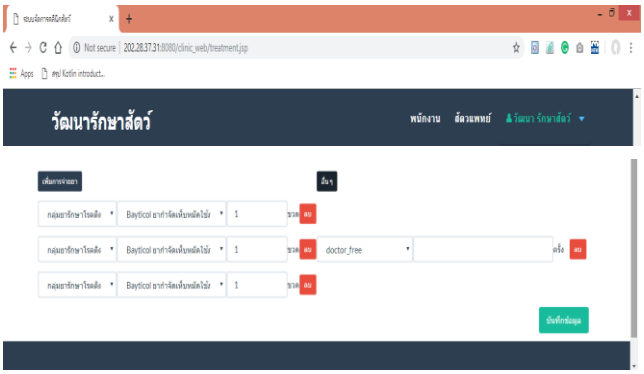

Fig. 8. The veterinarian dispenses medicines.

"Fig. 9" The manager inputs the type of report and date that he wants to look at.

Authorized licensed use limited to: University of Exeter. Downloaded on July 02,2020 at 16:40:58 UTC from IEEE Xplore. Restrictions apply.

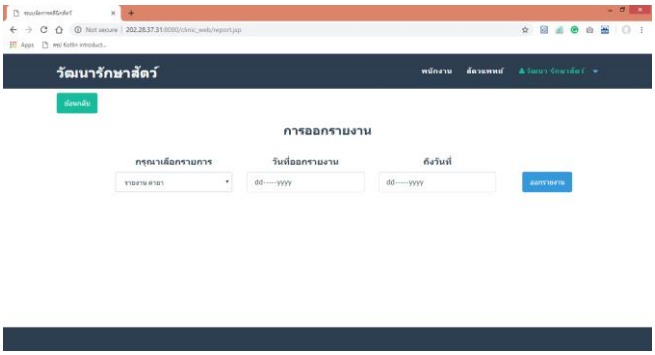

Fig. 9. The manager looks at the summary of income.

F. Evaluation

The EVM was set up and implemented by Watana Rak Sa Sat Veterinary clinic. The 100 client samples were randomly selected by simple random sampling technique. In first 50 cases of those were searched by manual registration for pet medical records and the last 50 cases were searched for registration by the EVM. We started the timer when the clients had begun to tell the name of themselves or pets until we got their medical records.

III. RESULTS

Comparison of access times between manual registration searching and EVM searching are shown in table II.

TABLE II. COMPARISON OF SEARCH TIME

	n	$\bar{x}$ (sec.)	s.d.
Manually registration search	50	37.09	23.24
EVM search	50	25.14	10.02

IV. CONCLUSION

The average EVM search time for getting pet medical record is less than the manual search time. While as the standard deviation value of EVM system is less than the manual searching also. That can conclude the EVM searching is more rapid than the manual process. But the both search times were include the time that the clients told the name of pet's owner that means sometime they could not remember whose correctly name.

ACKNOWLEDGMENT

The authors gratefully acknowledge the funding received towards our research from the Ministry of Science and Technology. We are thankful to Faculty of Science for providing infrastructure facilities for carrying out the work.

REFERENCES

[1] Ashish Kumar Tewari, Rashi, GulshanWadhwa, Sanjeev Kumar Sharma, and Chakresh Kumar Jain, "BIRS – Bioterrorism information retrieval system", Bioinformation, vol. 9(2), pp. 112–115, January 2013.

[2] A. Noraziah, Noor Naimah, and M. Nasratun, "Pet Shop Management System for Klinik Veterinar & Surgeri Jawhari", IEEE Xplore. Malaysia, Semtember 2008 [2008 International Symposium on Information Technology. Kuala Lumpur. Malaysia, August 2008].

[3] Cécile Cornou and Anders Ringgaard Kristensen, "Use of information from monitoring and decision support systems in pig production: Collection, applications and expected benefits", Livestock Science, vol. 157, pp. 552-567, November 2013.

[4] Chao-Hsi Huang, Pin-Yin Shen, and Yueh-Cheng Huang, "IoT-based physiological and environmental monitoring system in animal shelter", IEEE Xplore. Japan, August 2015 [2015 Seventh International Conference on Ubiquitous and Future Networks. Sapporo. Japan, July 2015].

[5] Chutchada Nusai, Sirisak Cheechang, Somkid Chaiphech, and Goragot Thanimkan, "Swine-vet : a web-based expert system of swine disease diagnosis", Procedia Computer Science, vol. 63, pp. 366-375, 2015.

[6] Emma Hooijberg, Ernst Leidinger, and Kathleen P. Freeman, "An error management system in a veterinary clinical laboratory", Journal of Veterinary Diagnostic Investigation, vol. 24(3), pp. 458-468, May 2012.

[7] Fu Zetian, Xu Feng, Zhou Yun, and Zhang Xiao Shuan, "Pig-vet: a web-based expert system for pig disease diagnosis", Expert Systems with Applications, vol. 29(1), pp. 93-103, July 2005.

[8] Gloria Ejehiohen Iyawa, Marlien Herselman, and Adele Botha, "Digital health innovation ecosystems: from systematic literaturereview to conceptual framework", Procedia Computer Science, vol. 100, pp. 244-252, 2016.

[9] Harold G. J. Bok, "Competency-based veterinary education: an integrative approach to learning and assessment in the clinical workplace", Perspect Med Educ, vol. 4(2), pp. 86-89, April 2015.

[10] Hassan A. Aziz, "A review of the role of public health informatics in healthcare", Journal of Taibah University Medical Sciences, vol. 12(1), pp. 78-81, February 2017.

[11] H.R. Meena and Y.P. Singh, "Importance of information and communication technology tools among livestock farmers: A review", Scientific Journal of Pure and Applied Sciences, vol. 2(2), pp. 1-9, 2013.

[12] Julie Jones-Diette, Natalie J. Robinson, Malcolm Cobb, Marnie L. Brennan, and Rachel S. Dean, "Accuracy of the electronic patient record in a first opinion veterinary practice", Preventive Veterinary Medicine, vol. 148, pp. 121-126, December 2017.

[13] Krone LM, Brown CM, and Lindenmayer JM, "Survey of electronic veterinary medical record adoption and use by independent small animal veterinary medical practices in Massachusetts", Journal of the American Veterinary Medical Association, vol. 245(3), pp. 324-332, August 2014.

[14] LaVallee E, Mueller MK, and McCobb E, "A systematic review of the literature addressing veterinary care for underserved communities", Journal of applied animal welfare science, vol. 20(4), pp. 381-394, 2017.

[15] Mehmet Bilen, Ali Hakan Işık, and Tuncay Yiğit, "Expert system software for domestic animals", IEEE Xplore. Turkey, November 2017 [2017 International Conference on Computer Science and Engineering (UBMK). Antalya. Turkey, October 2017].

[16] Natalie J. Robinson, Marnie L. Brennan, Malcolm Cobb, and Rachel S. Dean, "Agreement between veterinary patient data collected from different sources", The Veterinary Journal, vol. 205(1), pp. 104-106, July 2015.

[17] Revere D, Turner AM, Madhavan A, Rambo N, Bugni PF, Kimball A, and Fuller SS, "Understanding the information needs of public health practitioners: A literature review to inform design of an interactive digital knowledge management system", Journal of Biomedical Informatics, vol. 40(4), pp. 410-421, January 2007.

[18] Shahaji Phand, Rupasi Tiwari, and M. C. Sharma, "An information technology enabled animal health information system (AHIS) : Perceptions of dairy owners", Indian Journal of Animal Sciences, vol. 83(1), pp. 91-95, January 2013.

Authorized licensed use limited to: University of Exeter. Downloaded on July 02,2020 at 16:40:58 UTC from IEEE Xplore. Restrictions apply.